

Fruit Grading To Assist Selection of Fresh Fruits

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Abstract—India produces annually 10.9 % of the total fruit production in the world accounting for the second largest fruit producer in the world. But due to sudden weather changes and various other factors, the produced fruits end up reaching customers in a damaged state. In order for the customers to buy fresh fruits without physically handling them and in order to help a section of people in particular, image classification paves a way by identifying fruits as fresh and rotten based on their physical appearance.

Keywords—Fruit grading, Image classification, Computer Vision, cv2, Keras

I. INTRODUCTION

India is mainly known for its production of a variety of fruits. Different types of physic-geological conditions make it suitable for cultivation of distinct fruits and vegetables. 90% of the total horticultural production accounts from fruits and vegetables. The rate of production has increased significantly by almost 2 to 3 times compared to the previous decade. According to the reports from the final evaluations done in the year 2014–2015, the total production has expanded to around three times (2.87) compared to the rate in 1991–1992 and about twice (1.90) when compared with the generation in 2001–2002. The yearly increase in the demand of fruits and vegetables can be analyzed using the following statistics report as shown in Figure 1. The increasing demands of the nation for the fruits and the vegetables are supported by the contributions from various states of India. The statistics depicted in Figure 2 shows the percentage of production share of leading fruit producing States.

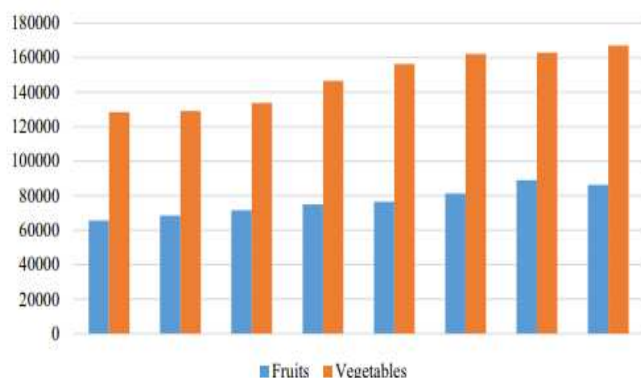


Fig 1. Fruits and Vegetables Production in India [1]

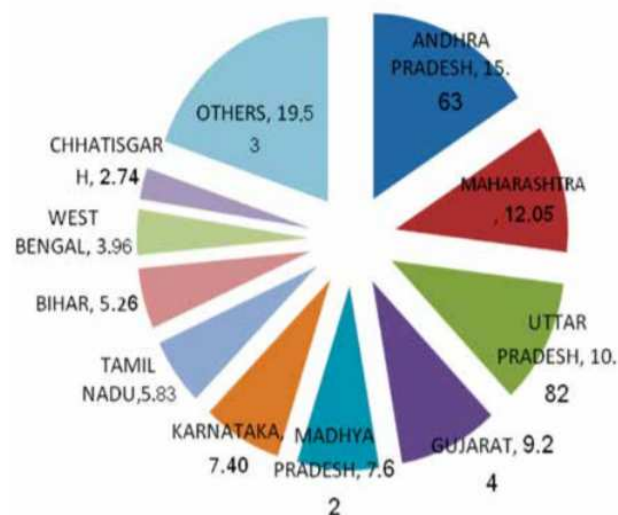


Fig 2. Production Share of Leading Fruit Producing States for 2017-18 [2]

India is majorly involved in the production of fruits and it can be seen that the rate of production and exports are increasing enormously. The rate of production and exports can be analyzed using the reports shown in Figure 3. Figure 4 shows the value of fruit exports from India as per the survey conducted for the FY 2015-2020.

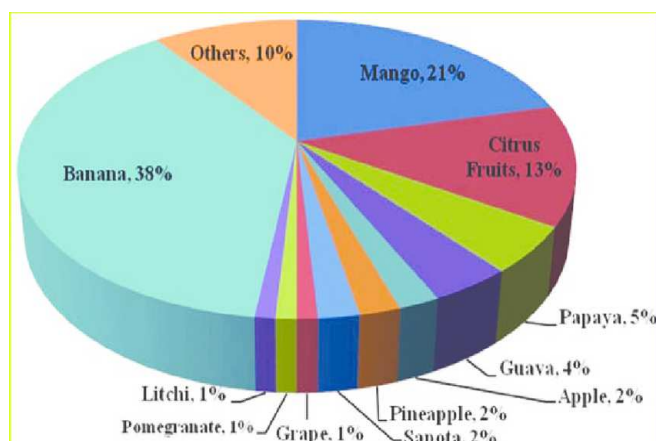


Fig 3. Production share of major fruits in India (2009-10) [3]

All data taken from the latest survey conducted.

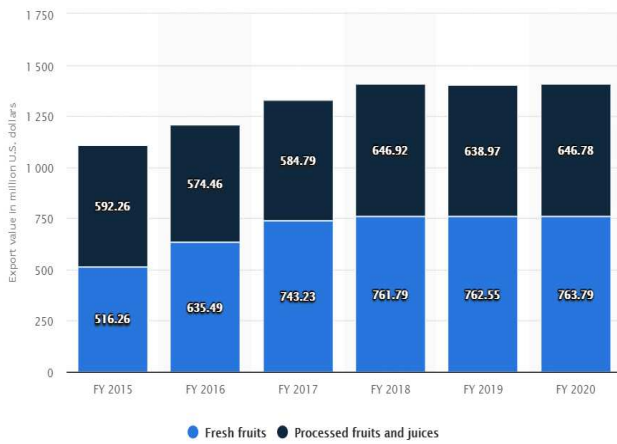


Fig 4. Value of fruit exports from India FY 2015-2020 [4]

But due to geographic and human interventions it sometimes leads to degradation in the quality of the production. Attack of pests such as locust swarms affect the quality of the produced fruits. Due to them being downgraded in their quality, these are not suitable for consumption. Even though these rotten fruits are not suitable for sale, they end up being sold in the market. The rotten fruits are not fit for consumption as they can affect human health, and also end up being a loss to the consumers who buy them. The overall loss from these factors can be well understood using the following report.

Section I contains the Introduction and Motivation for the paper, Section II contains Identification of fruits and grading of the fruits, Section III contains information regarding dataset as well as the information about the technique used and Working with dataset, Section IV contains Findings and future direction, Section V consists of Innovation in real time usage, Section VI Pros and cons, Section VII contains Conclusion and Section VIII contains References.

II. IDENTIFICATION AND GRADING OF THE FRUITS

Fruits have a varied physical appearance. Each fruit differentiates in its resemblance from the other. There are more than 2000 types of fruits around the world, each being physically different from each other. Each fruit has its own identification properties. With the increased development of the fruit industry, the use of effective technical methods to classify all kinds of fruits is a general trend. Computer vision system for fruit identification and disease identification of fruit is one among the current topics that is under research in the agriculture industry and is showing promising results. There are numerous pattern matching techniques that are used in order to achieve the same. The pattern matching techniques make use of complex computers and mathematical theories for producing results.

Mustafa et al., [5] in the year 2011 proposed a model that made use of the morphological features of the fruits, Factors such as major-axis, area, standard deviation red, minor axis and colour features such as mean red, mean green, mean blue, standard deviation green, standard deviation blue, mean hue, standard deviation hue, standard deviation saturation, mean saturation, mean intensity, standard deviation saturation were taken in consideration.

The model made use of five varieties of fruits, i.e., Oranges, Bananas, Apples, Carrots and Mangoes. The samples were collected by a digital camera and made use of for recognizable proof of fruits dependent on its morphological and colour feature that was achieved using the Probabilistic Neural Network (PNN) classifier. The demonstration resulted in a model with almost 90% accuracy. spelling and grammar.

Dubey et al. [5] (2012), proposed his methodology for the classification with respect to the species and the variety. The segmentation was accomplished using K-Mean clustering. The classification of fruits and vegetables was achieved using the method of multi-class SVM, which is based on the Improved Sum and Difference Histogram (ISADH) texture feature, which is a modified version of sum and difference histograms for texture classification as proposed by Unser in 1986. The ISADH is a texture feature, that is completely based on the total and contrast of intensity values of neighboring pixels of an image. The proposed approach had a much better efficiency than any of the previous models, thus implying that the contribution of ISADH feature for the purpose of classification is better than any other texture feature such as Global Color Histogram (GCH) and Color Coherence Vector (CCV). Various models were demonstrated for the classification of fruits based on varying factors and had different resulting accuracies. Table 1 summarizes the Analysis of Methodologies in Identification of Fruits.

A. Grading of Fruits

Grading includes classification of different fruits based on their defects and contamination as fresh and rotten. Assessing the quality of the fruits using grading helps consumers to choose the right fruits and vegetables. Grading has increased in recent years. Grading of fruits contains mainly the process of categorization of fruits based on three parameters: disease, defects in the fruits, and contamination on fruits. Usually manual grading has been carried out for a long time in order to ensure that the fruits are of the right quality for human consumption but it is a time consuming process and is unsuitable for visually impaired people. Thus considering all these issues, there is a need for an automated fruit grading system. The most reliable approach in order to achieve the same would be making use of the image processing systems for separation of lower grade fruits from the rest of the fruits.

There are various classification algorithms that have been taken into consideration in order to classify the fruits. Popular classification algorithms such as ANN, SVM, Naive Bayes algorithm etc to classify Apples, Oranges, Bananas, Dates and various other fruits. For example, Khojastehnazhand et al, in the year 2010, developed an algorithm that can be used to grade lemons. It consisted of two cameras. It takes in the input as the samples of different grades of lemons and are calibrated off-line. Later HSI color values and fruit volume are stored in a database. So with new inputs, values are compared to database values and the final grades of the respective fruits can be determined. The same algorithm can also be extended with various fruits. Zheng et al. [5] in 2012, came up with a new method that could detect the degree of browning of the mango fruit which was based on the least-squares support vector machine classifier. The degree can be identified using the L^*a^*b values and the Fractal Dimension(FD). For the LS SVM models the right classification rates of 85.19% and 88.89% were accomplished based on the FD($\gamma=1.13$, $\sigma^2=5.52$) and on $L^*a^*b^*$ ($\gamma=6.68$, $\sigma^2=2.44$). Various other

algorithms can be taken into consideration for the effective grading of fruits. Table 2 provides a Comparative analysis of different techniques applied for Classification of fruits

Table 1. Analysis of Methodologies in Identification of Fruits [5]

References	Fruit(s)	Segmentation	Features	Classification	Accuracy score (%)
Mustafa et al. (2011)	Apples, bananas, mangoes, carrots and oranges	N/A	Colour and geometrical feature	PNN	90
Dubey et al. (2012)	Plum, potato, cashew, onion, orange, kiwi, apple, watermelon, pear and peach	K-Mean clustering	Texture features	SVM	99
Savakar (2012)	Apple, mango, sweet lemon, chikoo and orange	N/A	Colour & texture features	ANN	94
Zawbaa et al. (2014)	Apples, oranges and strawberries	N/A	Shape, colour & SIFT feature	RF	96.97
Zhang et al. (2014)	Apples, bananas, plantains, tangerines, avocados, watermelons, cantaloupes, pineapple, pears	Split-and-merge algorithm	Texture & shape features	FNN	89.1
Khune et al. (2016)	Apples, watermelons, strawberries and lemons	N/A	Colour, size and shape features	SVM	N/A
Lu et al. (2016)	Pineapples, green plantains, cantaloupes, passion fruits, tangerines, apples, pears, grapes, strawberries, bananas, apples, pears, berries, watermelons	N/A	N/A	SLFN	89.5
Khaing et al. (2018)	Apples, bananas, blueberries, kiwi fruits, Raspberries, etc.	N/A	N/A	CNN	94

Table 2. Comparative analysis of different techniques applied for Classification of fruits

References	Fruit(s)	Segmentation	Features	Classification	Accuracy score
TONGUÇ et al. (2009)	Apples	N/A	Size & colour features	ANN	80%
Suresha et al. (2012)	Red apples & green apples	Threshold-based approach	Statistical features	SVM	100%
Pourjafar et al. (2013)	Mozafati dates	N/A	Geometric Features	ANFIS	93%
Hazra et al. (2013)	Tomatoes	N/A	Morphological shape-based features	Naive Bayes algorithm	100%
Alipasandi et al. (2013)	Peaches	N/A	Colour spaces and shape features	Neural Network	94.1% for mature and 98.5% for immature
Muhammad (2014)	Dates	N/A	Colour, texture, and shape features	SVM	99.67%
Kumar et al. (2015)	Oranges, sweet-limes and lemons	N/A	Colour	N/A	N/A
Samajpati et al. (2016)	Apples	K-Mean clustering	Colour & texture features	RF classifier	N/A
Mulyani et al. (2017)	Fuji apples	N/A	N/A	Fuzzy	85.71%
Rachmawati et al. (2017)	Apples, bananas, lemons, limes, oranges, peaches, and pears	N/A	Colour, hybrid & 3D shape features	Hierarchical classification	N/A
Choi et al. (2018)	Pears	N/A	Appearance & internal features	ANN	97.4%
Mim et al. (2018)	Himsagor mangoes	Global threshold value	Image features	Decision tree	96%
Oo et al. (2018)	Strawberry	N/A	Shape & size features	3-layer neural network	97%
Sidehabi et al. (2018)	Passion fruits	K-Mean clustering	RGB & A features	ANN	90%

III. FINDINGS AND FUTURE DIRECTION

The dataset[6] is chosen from Kaggle. It consists of 6 directories each in the train and the test set of data. The train data consists a total of **10901** images. Each directory is

labelled as one of freshapples, freshbanana, freshoranges, rottenapples, rottenbanana and rottenoranges. The test data consists of **2698** images.

After testing on the *test_data* it was found that the images were classified with an approximate accuracy of **98%** as

shown in Figure 5. Figure 6 and figure 7 depicts the loss function and accuracy obtained from the model respectively.

```
model.evaluate(X_test,Y_test)
```

```
85/85 [=====] - 17s 199ms/step - loss: 0.0552 - accuracy: 0.982  
[0.05517240986227989, 0.9822090268135071]
```

Fig 5. Model Evaluation

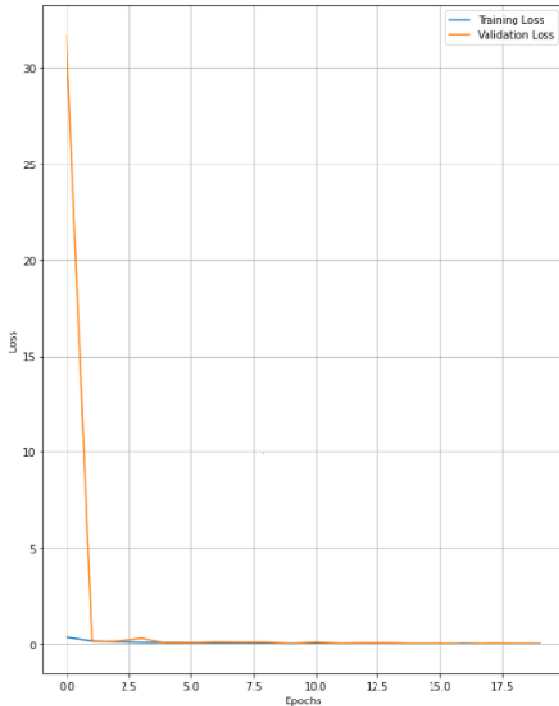


Fig 6. Loss graph

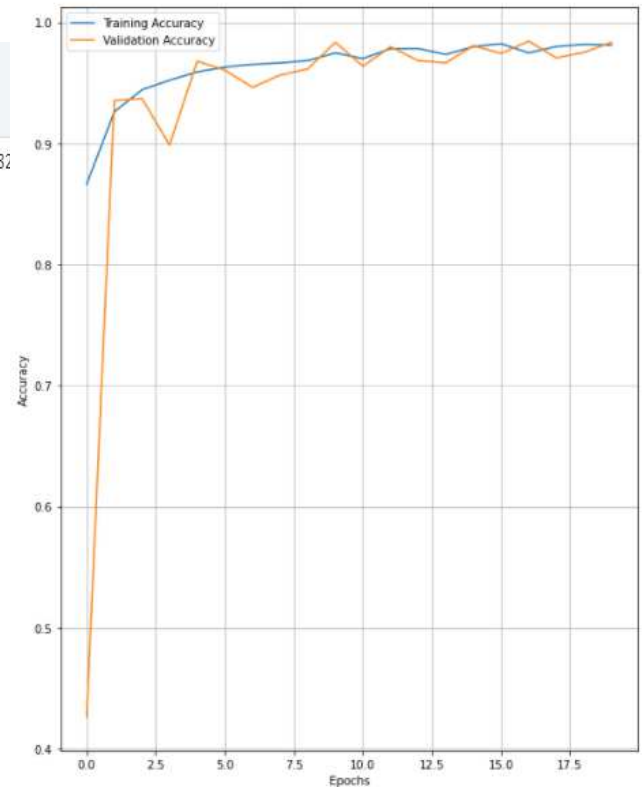
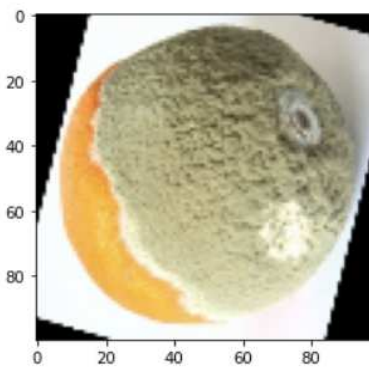


Fig 7. Accuracy graph

Hence, this model can be effectively used to grade the fruits based on their physical appearance with great accuracy. Figure 6 shows the prediction of grade for the sample input image of fruit.

```
plt.imshow(X_test[1])
```

```
<matplotlib.image.AxesImage at 0x7f53b49d2210>
```



```
model.predict(X_test[1].reshape(1,100,100,3))
```

```
array([[1.]], dtype=float32)
```

Fig 8. Prediction of Grade for the Given Image

This experiment can be extended to train a model to grade a variety of fruits and vegetables.

Further, this can also be used to detect tampered packets or damaged items when trained with the particular dataset.

Hence, it would be effectively used to detect damages without physically handling the products.

IV. INNOVATION IN REAL TIME USAGE

The result of the model can be used in various places in order for the welfare of mankind. One of the major places where the results can be used in the best possible way is inculcating it in the equipment in order to extend a helping hand for the visually impaired people in order to ease the process of selection of the fruits. It can be clubbed with the olfactory senses of the individual which are generally put in use, combined with this can successfully help them select the right quality of fruits. The whole model can be narrowed down by fitting it in a form of glass for better portability and ease of use. The glasses capture simultaneous sets of images and analyze them based on the results from the trained model. The input follows a two-step process wherein the first step is the Identification of the fruit and the second step is the grading of the fruit. The input data is compared with the data in the database and results as an output describing the quality of the fruits. The glasses are built in with a set of speakers that can directly deliver the test results to the user in the form of an audio.

The pair of glasses can be trained further using the continuous input from the multiple users and thus resulting in providing efficient results to a greater extent. It is a great tool that can act as an assistance for the visually impaired people and a step taken in the direction to make things simpler for them.

V. ADVANTAGES AND DISADVANTAGES

The advantages include no physical contact with the item. This prevents accidental damage and also prevents spreading of contaminants such as bacteria and viruses.

The model provides almost perfect results; hence it is reliable for further use in manufacturing units and even retail stores where the model can be integrated with the CCTV

cameras to check for damaged products. The model can be trained to recognize damage in the sensitive parts of a product such as a glass part, and hence can check if a product is damaged.

The disadvantage is that it is impossible to predict accurately without capturing images from all directions since a fruit may look fresh from one angle but may be damaged on the other. Hence, it is a tedious task to accurately grade it without compromising on speed or on complexity of systems used in places which deal with tons of fruits and vegetables.

VI. CONCLUSION

To conclude, an effective model to grade fruits and vegetables which can be extended to even products sold in packets or boxes is necessary in order to avoid physical handling of items. Sophisticated hardware such as large memory and multiple cameras may be necessary in places handling huge volumes of these items. But overall, for usage in places requiring lesser scanning of images, the process takes seconds to finish with simple equipment such as an app on a mobile phone.

VII. REFERENCES

- [1] Horticulture statistics
<https://agricoop.nic.in/sites/default/files/Horticulture%20Statistics%20at%20a%20Glance-2018.pdf>
- [2] Fruit and vegetable production statistics
https://www.researchgate.net/figure/Fruits-and-Vegetables-Production-in-India_fig2_279866746
- [3] Share of production of various fruits
https://www.researchgate.net/figure/Production-share-of-major-fruits-in-India-2009-10-Source-Indian-Horticulture-Database_fig1_274085550
- [4] Export values of various fruits
<https://www.statista.com/statistics/652191/export-value-of-fruits-by-type-india/>
- [5] Behera, Santi Kumari, et al. "Identification, classification & grading of fruits using machine learning & computer intelligence: a review." *Journal of Ambient Intelligence and Humanized Computing* (2020): 1-11
- [6] <https://www.kaggle.com/toandominh/final-project-nhom2-chieu2-tiet10-12>